

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

CO-4 AT-1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Annexure A

Data Interpretation

Criteria	Understanding of Data (2.5)	Analysis (2.5)	Application of Concepts (2)	Conclusion (1.5)	Clarity (1)	Total(10)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

192372163

I S. chandra sheela (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S. chandra sheela

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K SSE

1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Q2. Dataset: E-Commerce Order Analysis

Order ID Product Category Order Value (₹)

O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

Q3. Dataset: Mobile App User Behavior

	User Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Q4. Dataset: Agricultural Crop Monitoring

	Farm Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Q5. Dataset: Employee Performance Evaluation

	Employee Productivity Score	Attendance (%)	Skill Assessment Score
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model Score

K=2 0.48

K=3 0.68

K=4 0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

① Data Set: Hospital Patients Segmentation

Patient ID	Age	Expense (₹) Monthly medical	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values:

<u>k</u>	<u>SSE</u>
1	600
2	350
3	220
4	200
5	190

Decrease in SSE

$$600 - 350 = 250$$

$$350 - 220 = 130$$

$$220 - 200 = 20$$

$$200 - 190 = 10$$

Elbow Calculation

- from $k=1 \rightarrow k=2$, sse decrease by 250
- from $k=2 \rightarrow k=3$, sse decrease by 180
- from $k=3 \rightarrow k=4$, sse decrease by 20
- from $k=4 \rightarrow k=5$, sse decrease by only 20

The major reduction stops after $k=3$

$\therefore$ Elbow point = $k=3$

Optimal Number of cluster = 3

② Dataset : E-Commerce Order Analysis

Order-ID	Product Category	Order value
1	Electronics	55000
2	clothing	25000
3	Electronics	57000
4	furniture	30000
5	clothing	27000

sol

Given that

Orders:

- Electronics : 1 = 55,000 , 3 = 57,000
- clothing : 2 = 25000 , 5 = 27000
- furniture : 4 = 30,000

The Most Similar order are

01 and 3

- Both are electronics
- order values are very close:

$$57,000 - 55,000 = 2,000$$

2 and 3

- Both are clothing
- Order values are close:

$$27,000 - 25,000 = 2,000$$

④ (furniture, 30,000) is relatively close in value to the clothing orders, but its category is different

2. cluster formation

The dendrogram would first merge most similar orders

cluster 1:

$\{0, 3\}$

Electronics with high order values:

cluster 2:

$\{2, 8\}$

Clothing with medium order values:

cluster 3:

$\{4\}$

Furniture with order value of 30,000

2. on appropriate interpretation is a cluster

cluster	orders	category	order value pattern
C1	9, 3	electronics	45,000 - 75,000 → High
C2	2, 5	clothing	25,000 - 35,000 → medium
C3	4	furniture	80,000

Conclusion :
The dendrogram indicates that 3 clusters provide meaningful segmentation.
 $C1 = \{2, 3, 9\}$, $C2 = \{2, 5\}$, $C3 = \{4\}$

⑧ Data set: Mobile App user Behaviour

User	Daily usage Time	Login Frequency	In-App purchase
U1	100	18	2500
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	120	20	1800

⑧ The users can be divided into approximately 2 behavioural clusters

2. cluster formation

cluster 1 - highly active users

$C1 = \{U1, U3, U5\}$

These users have:

- High daily usage : 120 - 140 minutes
- High login frequency : 18 - 22
- Relatively low purchase : 1500 - 2500

4. Soft clustering Vs Hard clustering

Hard clustering

Each user belongs to only one cluster

example

$U_2 \rightarrow C_2$

Soft clustering

A user can have different membership probabilities in different clusters

Example

$U_2 \rightarrow 85\% C_2 \text{ and } 15\% C_1$

Advantages of Soft clustering

- Handles overlapping behavior:

A customer may show characteristics of more than one segment

- Better customer understanding

membership probabilities indicates how strongly a user belongs to a different group

- improve decision-making

The company can identify user who are borderline b/w segments and target them differently.

They are highly engaged but have lower spending

Cluster 2 - high - spending, low active user

0.20 250,000

These users have

- low daily usage: 50-60 minutes
- log in frequency: 4-5
- high purchase: £5,000 - £10,000

They are less active but generate higher revenue

3. Probability - based Membership

In both clustering, membership can be represented using probability

for example	Active users	high - spending cluster
0.1	0.90	0.10
0.2	0.15	0.05
0.3	0.95	0.95
0.4	0.05	
0.5	0.00	

It $U2=0.85$ for the high - spending cluster, it means U2 has 85% membership probability in that cluster

Agricultural crop Monitoring

Farm	Rainfall	Soil fertility	Score yield crop (tons)
F1	300	8	85
F2	500	4	
F3	400		70
F4	280	6	80
F5	520	9	90
		3	65

The main issue is that three attributes have different ranges:

- Rainfall : 280 - 520
- Soil fertility : 3 - 9
- Crop yield : 65 - 90

∴ rainfall can dominate the clustering distance if normalization is not applied

1. Before Normalization

$$d = \sqrt{(300 - 280)^2 + (8 - 9)^2 + (85 - 90)^2}$$

$$= \sqrt{20^2 + (-1)^2 + (-5)^2}$$

$$= \sqrt{400 + 1 + 25}$$

$$= \sqrt{426}$$

$$\boxed{d \approx 20.64}$$

2. Apply min-max Normalization

$$y' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Rainfall

minimum = 280, maximum = 520

For F_1 :

$$F_1 = \frac{300 - 280}{520 - 280} = \frac{20}{240} = 0.083$$

$$F_4 = \frac{280 - 280}{520 - 280} = 0$$

Soil Fertility

Min = 8, max = 9

$$F_1 = \frac{8 - 3}{9 - 3} = \frac{5}{6} = 0.833$$

$$F_4 = \frac{9 - 3}{9 - 3} = 1$$

Crop yield

min = 65, max = 90

$$F_1 = \frac{85 - 65}{90 - 65} = \frac{20}{25} = 0.8$$

$$F_4 = \frac{90 - 65}{90 - 65} = 1$$

$$S_0 = F_1 (0.083, 0.833, 0.8)$$

$$F_4 = (0.1, 1)$$

Distance After Normalization

$$d = \sqrt{(0.083 - 0)^2 + (0.833 - 1)^2 + (0.8 - 1)^2}$$
$$= \sqrt{0.0069 + 0.0279 + 0.04}$$
$$= \sqrt{0.0748}$$

$$d = 0.274$$

4. Expected cluster

Cluster 1: High Fertility and high yield

$\{F_1, F_4\}$

Cluster 2: High rainfall, low fertility and low yield

$\{F_2, F_5\}$

Cluster 3: moderate condition

$\{F_3\}$

⑤ Data Set : Employee performance Evaluation

Employee	productivity	Score Attendance	Skill Assessment Score
E1	88	92	
E2	62	75	84
E3	78	88	68
E4	55	65	72
E5	93	96	58
			90

Silhouette Scores

$k=2$ 0.48

$k=3$ 0.68

$k=4$ 0.52

Sol The Silhouette Score measures how well each employee fits within its assigned cluster compared with other clusters.

The score ranges from -1 to 1

- closer to +1 $\rightarrow$ well separated and meaningful cluster.
- Around 0 $\rightarrow$ overlapping cluster
- Negative $\rightarrow$ employee may be assigned to wrong cluster

1. Analyze the Given Score

No of cluster	Silhouette score	Interpretation
$k=2$	0.48	moderate clustering
$k=3$	0.68	strong clustering
$k=4$	0.52	moderate clustering

Comparison

$$0.48 < 0.68 < 0.52$$

The highest Silhouette score is 0.68 for $k=3$

2. Optimal No of cluster

$$\therefore k=3$$

is the most suitable clustering model

A score of 0.68 indicates that employees are reasonably well separated and employees within each group have similar performance.

3. possible Employee Segmentation

Based on given score, employees can be interpreted into three performance groups

Cluster 1: High performer

1. High productivity
2. High attendance
3. High skill score

{E1, E5}

Cluster 2: Average performer

- Moderate productivity
- Good/moderate attendance
- Moderate skill level

{E3}

Cluster 3:-

- Low productivity
- Low attendance
- Lower skill score

{E2, E4}

4. Why k=3 is Better

If $k=2$ the score is only 0.48, meaning some different type of employees may be combined into same group.

If $k=3$ the score increases to 0.68, indicating better separation b/w high average and low performing employees.

If $k=4$ the score falls to 0.52. This suggests that creating four groups does not improve the segmentation and may be cluster unnecessarily fragmented.

5. Employee performance evaluation

using it to help company

- identify high performing employees for rewards and promotions
- identify high average performance who may benefit from additional

training

- identify employee requiring performance improvement program
- compare productivity attendance and skill together.